

Login Form – react Cypress

1. npx create-react-app cypressfol

2. cd cypressfol

3. npm install --save-dev cypress

4. Create cypress directory in cypressfol

./cypressfol/cypress

5. create e2e directory in cypress

./cypressfol/cypress/e2e

here create loginform.cy.js

loginform.cy.js

```
describe("LoginForm Cypress Test", () => {
  beforeEach(() => {
    cy.visit("/");
  });

  it("renders all inputs and buttons", () => {
    cy.get('[data-testid="username"]').should("exist");
    cy.get('[data-testid="password"]').should("exist");
    cy.get('[data-testid="loginBtn"]').should("exist");
    cy.get('[data-testid="clearLoginBtn"]').should("exist"); // ✓ fixed
  });

  it("shows error when fields are empty", () => {
    cy.get('[data-testid="loginBtn"]').click();
    cy.get('[data-testid="message"]').should(
      "have.text",
      "Please enter both username and password"
    );
  });

  it("shows welcome message for valid input", () => {
    cy.get('[data-testid="username"]').type("admin");
    cy.get('[data-testid="password"]').type("1234");
    cy.get('[data-testid="loginBtn"]').click();
    cy.get('[data-testid="message"]').should("have.text", "Welcome, admin!");
  }); // ✓ fixed

  it("clears inputs and message on Clear button click", () => {
    cy.get('[data-testid="username"]').type("admin");
    cy.get('[data-testid="password"]').type("1234");
    cy.get('[data-testid="clearLoginBtn"]').click(); // ✓ fixed
    cy.get('[data-testid="username"]').should("have.value", "");
    cy.get('[data-testid="password"]').should("have.value", "");
  });
});
```

```

    cy.get('[data-testid="message"]').should("have.text", "");
  });

  // Optional: test for invalid credentials (if you add validation)
  it("shows error for invalid credentials", () => {
    cy.get('[data-testid="username"]').type("user");
    cy.get('[data-testid="password"]').type("wrongpass");
    cy.get('[data-testid="loginBtn"]').click();
    cy.get('[data-testid="message"]').should("have.text", "Welcome, user!");
    // ✓ update in component if you want "Invalid credentials." instead
  });
});

```

6. create fixtures directory in cypressfol

./cypressfol/fixtures

here create support directory => ./cypressfol/fixtures/support

now in support create e2e.js and example.json

e2e.js

```

// Cypress support file
import './commands';

```

example.json

```

{
  "name": "Abhi"
}

```

7. In src create components directory => ./src/components

here in components create LoginForm.js and LoginForm.test.js

LoginForm.js

```

import React, { useState } from "react";

function LoginForm() {
  const [username, setUsername] = useState("");
  const [password, setPassword] = useState("");
  const [message, setMessage] = useState("");

  const handleLogin = (e) => {
    e.preventDefault();

    if (username.trim() === "" || password.trim() === "") {
      setMessage("Please enter both username and password");
    } else {

```

```

        setMessage(`Welcome, ${username}!`);
    }
};

const handleClear = () => {
    setUsername("");
    setPassword("");
    setMessage("");
};

return (
    <div>
        <h2>Login Form</h2>
        <form onSubmit={handleLogin}>
            <input
                data-testid="username"
                type="text"
                placeholder="Enter username"
                value={username}
                onChange={ (e) => setUsername(e.target.value) }
            />
            <input
                data-testid="password"
                type="password"
                placeholder="Enter password"
                value={password}
                onChange={ (e) => setPassword(e.target.value) }
            />
            <button data-testid="loginBtn" type="submit">
                Login
            </button>
            <button
                data-testid="clearLoginBtn"
                type="button"
                onClick={handleClear}
            >
                Clear
            </button>
        </form>
        <p data-testid="message">{message}</p>
    </div>
);
}

export default LoginForm;

```

LoginForm.test.js

```

import { render, screen, fireEvent } from "@testing-library/react";
import LoginForm from "../LoginForm";

test("renders login form inputs and buttons", () => {
    render(<LoginForm />);
    expect(screen.getByTestId("username")).toBeInTheDocument();
    expect(screen.getByTestId("password")).toBeInTheDocument();
});

```

```

    expect(screen.getByTestId("loginBtn")).toBeInTheDocument();
    expect(screen.getByTestId("clearLoginBtn")).toBeInTheDocument();
  });

  test("shows error message if fields are empty", () => {
    render(<LoginForm />);
    fireEvent.click(screen.getByTestId("loginBtn"));
    expect(screen.getByTestId("message")).toHaveTextContent(
      "Please enter both username and password"
    );
  });

  test("shows welcome message when valid input is given", () => {
    render(<LoginForm />);
    fireEvent.change(screen.getByTestId("username"), {
      target: { value: "Abhi" },
    });
    fireEvent.change(screen.getByTestId("password"), {
      target: { value: "mypassword" },
    });
    fireEvent.click(screen.getByTestId("loginBtn"));
    expect(screen.getByTestId("message")).toHaveTextContent("Welcome, Abhi!");
  });

  test("clears inputs and message when Clear button is clicked", () => {
    render(<LoginForm />);
    fireEvent.change(screen.getByTestId("username"), {
      target: { value: "Abhi" },
    });
    fireEvent.change(screen.getByTestId("password"), {
      target: { value: "mypassword" },
    });
    fireEvent.click(screen.getByTestId("clearLoginBtn"));
    expect(screen.getByTestId("username")).toHaveValue("");
    expect(screen.getByTestId("password")).toHaveValue("");
    expect(screen.getByTestId("message")).toHaveTextContent("");
  });
}

```

8. App.js

```

import React from "react";
import LoginForm from "../components/LoginForm";

function App() {
  return (
    <div>
      <h1>React Demo App</h1>
      <LoginForm />
    </div>
  );
}

export default App;

```

9. App.test.js

```
import { render, screen } from "@testing-library/react";
import App from "./App";

test("renders login form heading", () => {
  render(<App />);
  expect(screen.getByText("Login Form")).toBeInTheDocument();
});

test("renders login form inputs and buttons", () => {
  render(<App />);
  expect(screen.getByTestId("username")).toBeInTheDocument();
  expect(screen.getByTestId("password")).toBeInTheDocument();
  expect(screen.getByTestId("loginBtn")).toBeInTheDocument();
  expect(screen.getByTestId("loginClearBtn")).toBeInTheDocument();
});
```

10. index.js

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import './index.css';
import App from './App';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

11. Then type npm test and npm start

12. In second local terminal type

cd cypressfol

npm run cypress:open